

Problem Set 2

Charley Baker

Voting for a PTA

Globalization has been in part due to technological advances (as argued in Problem Set 1), but also to a radical policy change after World War II. According to Ptashkina (2023) virtually all countries in the world are members of at least one Preferential Trade Agreement (PTA) such as free-trade agreements or custom unions of various sorts. As of 2021 a total of 354 PTAs were in force. The provisions of these agreements cover 80–90% of bilateral trade between signatories and apply to more than half of total world trade.¹

Here you are asked to be the advisor of the government of a SOE that is considering signing a PTA with a much larger economy (think of this large economy as the United States or the European Union). The signing of this PTA would effectively reduce transport costs from τ_H to τ_L , with $\tau_H > \tau_L$. The government is preparing a policy proposal that consists of (i) signing the PTA, and (ii) making a redistribution within the SOE. This policy proposal will be voted in a referendum. If more than half of the citizens of this SOE votes for it, the policy proposal will be implemented. Otherwise, the policy proposal will not be implemented. Citizens always vote for the alternative that offers them the highest welfare.

The government wants you to help prepare the policy proposal. To do so, you are asked to use the Heckscher-Ohlin model. Recall that, in this model, the welfare effects of the policy proposal depend on international commodity prices; factor endowments; and also on the ownership of these factor endowments. Regarding the latter, there are L citizens in this SOE, and all of them owns one unit of labor. The other production factor is a stock of capital equal to K , which is unevenly distributed, though a fraction $1 - x \in (0, 1)$ of the citizens do not own any capital. We refer to these citizens as the workers. The remaining fraction x of the citizens own all the capital in equal proportions, i.e., each of them owns a share $\frac{K}{xL}$ of the capital stock. We refer to these citizens as the capitalists (although they also own one unit of labor).

(Hint: This model is the same as the one we did in class, with capital K , instead of skilled workers H . But some people now own both factors and some do not.)

The government wants to sign the PTA, since it understands that the gains from trade are positive. The government wants to raise the welfare of workers as much as possible. You must design a redistribution policy that achieves these goals. The only restriction you face when designing this redistribution is that, after implementing it, the capitalists cannot be poorer than the workers. This restriction might be binding or not.²

In particular, you are asked to answer the following questions:

1. Would the PTA be approved without any redistribution?
2. What is the optimal redistribution from the point of view of the government?

¹Bibliographical reference: Ptashkina, M. (2023), “Revisiting the Effects of Preferential Trade Agreements,” Chapter 1, PhD dissertation.

²Where does this restriction come from? Perhaps the government is Rawlsian and it really cares for the poor and not for the workers specifically. Perhaps the capitalists will destroy their capital and become workers if this makes them better off. This would have disastrous effects on the economy as a whole.

Naturally, your answers depend on the specifics of the SOE which, as mentioned above, are given by international commodity prices, i.e., P_L and P_K ; factor endowments, i.e., L and K ; and the ownership of these factor endowments, i.e., x . Let W and R be factor prices, i.e., the wage and the rental rate of capital. To simplify the discussion, assume that the SOE is an exporter of L-goods. This means that the domestic prices of L- and K-goods before the PTA is signed are $\frac{P_L}{\tau_H}$ and $\tau_H P_K$. (Hint: Do not be afraid to make assumptions. It is always useful to study simple cases first where you can identify a clear answer, and move on later to more complex settings.)

1. Would the PTA be approved without any redistribution?

To see if the PTA would be approved, we compare total utility of the SOE before the agreement is signed and after, when transport costs decrease from τ_H to τ_L and trade increases. We also consider how the utility of different population segments (i.e. workers and capitalists) change. To find the impact of the PTA on utility, we set up our model and find our market-clearing conditions as follows:

We assume that there are 2 types of goods/industries: L-goods, which are labour-intensive, and K-goods, which are capital-intensive. We can express their production functions in the two equations below:

$$Q_L = L^{1-\alpha_L} K^{\alpha_L} \quad \text{and} \quad Q_K = L^{1-\alpha_K} K^{\alpha_K} \quad (1)$$

The SOE is modeled as a partial equilibrium environment in which international commodity prices are exogenous and taken as given. As such, we do not solve for goods prices as a function of production quantities and costs. Rather, we express factor prices, W and R , and industry productions, Q_L and Q_K , as a function of factor endowments, K and L and goods prices, $\frac{P_L}{\tau_i}$ and $\tau_i P_K$.

Given the Heckscher-Ohlin theorem and that the SOE is an exporter of L-goods, we assume that the SOE in question is labour-abundant:³

$$\frac{L_1}{K_1} > \frac{L_1 + L_2}{K_1 + K_2} > \frac{L_2}{K_2} \quad (2)$$

To find the income earned from both factor markets, we state the market clearing conditions for factor markets. These conditions follow assumptions related to our theory of production: unit-cost minimisation, perfect competition and the Cobb-Douglas production functions exhibiting constant returns to scale. The equations below state that the SOE's total income is the sum of the income earned by labourers and capitalists from producing L-goods and K-goods. Equation (3) says that total labour income is labour's share of the total sales of L- and K-goods:

$$WL = (1 - \alpha_L)P_L Q_L + (1 - \alpha_K)P_K Q_K \quad (3)$$

Capital income (equation (4)) is capital's share of the total sales of L- and K-goods:

$$RK = \alpha_L P_L Q_L + \alpha_K P_K Q_K \quad (4)$$

Note that labour income can be split for both population segments:

³We denote the SOE with subscript 1 and the large economy with subscript 2. When later analysing the impacts of the PTA, we drop subscripts as we only consider the impact on SOE variables.

$$WL = WxL + W(1-x)L \quad (5)$$

With these, we can also express income in terms of the two different population segments in the economy. For the workers, equation (6) represents their earnings which is just labour income:

$$W(1-x)L = (1-x)[(1-\alpha_L)P_LQ_L + (1-\alpha_K)P_KQ_K] \quad (6)$$

For the capitalists, their income is derived from labour wages, plus the return earned on capital:

$$WxL + RK = x[(1-\alpha_L)P_LQ_L + (1-\alpha_K)P_KQ_K] + \alpha_L P_LQ_L + \alpha_K P_KQ_K \quad (7)$$

We can then also find market-clearing conditions in the K- and L-industries where supply equals demand (where demand is comprised of both the SOE and abroad):

$$P_{L1}Q_{L1} + P_{L2}Q_{L2} = \mu_L(W_1L_1 + R_1K_1 + W_2L_2 + R_2K_2) \quad (8)$$

$$P_{K1}Q_{K1} + P_{K2}Q_{K2} = \mu_K(W_1L_1 + R_1K_1 + W_2L_2 + R_2K_2) \quad (9)$$

Equation (8) says the total values of sales of L-goods globally is equal to the total value of purchases of L-goods globally. Equivalently equation (9) represents the total values of sales and purchases for K-goods globally.

These conditions determine the welfare in our economy. We can express it as indirect utility in the general form given by earnings over the price level, see equation (10):

$$U_n = \frac{WL + RK}{P_L^{\mu_L} P_K^{\mu_K}} \quad (10)$$

In our model with adjusted prices for the SOE, this can be re-expressed as:⁴

$$U_1 = \frac{(1-\alpha_L)\frac{P_L}{\tau_H}Q_L + (1-\alpha_K)\tau_H P_KQ_K + \alpha_L\frac{P_L}{\tau_H}Q_L + \alpha_K\tau_H P_KQ_K}{(\frac{P_L}{\tau_H})^{\mu_L}(\tau_H P_K)^{\mu_K}} \quad (11)$$

The overall impact on utility depends on whether the SOE is specialised or diversified before the PTA and how the agreement changes its production position afterward. To conclude if the PTA will be approved without any redistribution, we shall analyse three cases, namely A, B and C.

Cases A, B and C

We define τ^* as the transport-cost threshold at which the SOE shifts from a diversified economy to specializing in L-goods. When transport costs exceed this threshold, the SOE produces both L- and K-goods, exporting L-goods and importing some K-goods. In this case, the domestic and import prices of K-goods are equal. This equality means the foreign country produces K-goods more cheaply, but transport costs offset its advantage. As transport costs fall, imported K-goods become cheaper even after accounting for τ , eventually making the foreign country the sole lowest-cost supplier. The SOE, constrained by its factor endowments, can no longer compete and therefore specialises in L-goods.

⁴We use equations (3) and (5) for this and we substituted the general price for the domestic price which is given in the problem.

Case A: Specialisation before and after the PTA

Under Case A, we assume that our SOE specialised in L-goods before the PTA: $\tau^* > \tau_H > \tau_L$. Following the PTA, the SOE remains specialised in L-goods production, as they are still the lowest-cost supplier of L-goods and lower transport costs make exporting more L-goods possible due to increased demand of L-goods from abroad. Output and exports of Q_L rise, increasing wages and overall earnings for all labour owners (workers and capitalists). Since the welfare of both workers and capitalists in the specialised case is improved through the lower τ , there is no incentive to diversify. The increase in earnings would be disproportionate for both capitalists and workers (as capitalists would also earn a return on the amount of K used in the production of L-goods), but nonetheless both incomes would increase.

Consequently, the utility of all agents of the economy increases as earnings rise, coupled with a fall in prices of goods from $\frac{P_L}{\tau_H}$ and $\tau_H P_K$ to $\frac{P_L}{\tau_L}$ and $\tau_L P_K$ respectively. So, the PTA would be approved, without a redistribution policy. This can be seen mathematically from equations (12) and (13).

Utility **before** the PTA:

$$U_1 = \frac{(1 - \alpha_L) \frac{P_L}{\tau_H} Q_L + \alpha_L \frac{P_L}{\tau_H} Q_L}{\left(\frac{P_L}{\tau_H}\right)^{\mu_L} (\tau_H P_K)^{\mu_K}} \quad (12)$$

Utility **after** the PTA:

$$U_1 = \frac{(1 - \alpha_L) \frac{P_L}{\tau_L} Q_L + \alpha_L \frac{P_L}{\tau_L} Q_L}{\left(\frac{P_L}{\tau_L}\right)^{\mu_L} (\tau_L P_K)^{\mu_K}} \quad (13)$$

Given that $\tau_H > \tau_L$, a reduction in transport costs via the preferential trade agreement leads to an increase in welfare, U_1 .

Case B: Diversified before and after the PTA

We now consider the case where the SOE is diversified and produces both goods. In order to compare the changes to welfare of capitalists and workers, we use the welfare equation for each segment of the population (workers and capitalists) rather than the combined welfare equation used in case A. The expression of our utility equation is also slightly different because we consider the payout of K-goods which are now produced in addition to the L-goods. This case is more ambiguous since not all agents are benefiting from this agreement. We have some winners and some losers.

In this case, $\tau_H > \tau_L > \tau^*$, i.e. the lower transport costs from the PTA are not high enough for the SOE to change from a diversified to a specialised economy. Thus the SOE is exporting L-goods and importing K-goods before and after the PTA, however at higher volumes post-PTA. Here the workers will enjoy higher wages due to the increase in demand for L-goods, Q_L , while the capitalists in the SOE lose due to an increase of imports of K-goods from abroad and a reduction in the production of K-goods domestically, Q_K . This is because we assume that the capitalists' income is more greatly impacted by the return to capital than the wage to labour ⁵.

⁵See the Stolper-Samuelson theorem: the increase in labour wage is proportionally higher than the reduction in return to capital in the production of L-goods, when the price for L-goods rises relative to K-goods

If we compare utility before and after the PTA (with equations (14) to (18) written below), we see that utility increases for workers as the income they gain from producing more labour-abundant L-goods offsets the income lost from the lower production of capital-intensive K-goods. For capitalists, they have lower earnings as the additional labour income from producing more L-goods does not offset the loss in returns to capital from a lower production of K-goods (owing to our assumption). Throughout the rest of our analysis (for cases B and C in questions 1 and 2), we assume that the fall in the price level does not offset the fall in income for capitalists.

The PTA will only be approved without redistribution when $(1 - x) > x$ i.e. the share of workers is higher than the share of capitalists (and strictly more than half need to vote for the PTA). If the inverse were true and $(1 - x) \leq x$, a greater share of the population would have a lower utility due to lower earnings, the PTA would not be signed.

The general utility equations for workers and capitalists:

$$U_W = \frac{(1 - x)WL}{P_L^{\mu_L} P_K^{\mu_K}} \quad U_C = \frac{WxL + RK}{P_L^{\mu_L} P_K^{\mu_K}} \quad (14)$$

Utility of workers and capitalists **before** PTA:

$$U_W = \frac{(1 - x)[(1 - \alpha_L)\frac{P_L}{\tau_H}Q_L + (1 - \alpha_K)\tau_H P_K Q_K]}{(\frac{P_L}{\tau_H})^{\mu_L} (\tau_H P_K)^{\mu_K}} \quad (15)$$

$$U_C = \frac{x[(1 - \alpha_L)\frac{P_L}{\tau_H}Q_L + (1 - \alpha_K)\tau_H P_K Q_K] + \alpha_L \frac{P_L}{\tau_H}Q_L + \alpha_K \tau_H P_K Q_K}{(\frac{P_L}{\tau_H})^{\mu_L} (\tau_H P_K)^{\mu_K}} \quad (16)$$

Utility of workers and capitalists **after** PTA:

$$U_W = \frac{(1 - x)[(1 - \alpha_L)\frac{P_L}{\tau_L}Q_L + (1 - \alpha_K)\tau_L P_K Q_K]}{(\frac{P_L}{\tau_L})^{\mu_L} (\tau_L P_K)^{\mu_K}} \quad (17)$$

$$U_C = \frac{x[(1 - \alpha_L)\frac{P_L}{\tau_L}Q_L + (1 - \alpha_K)\tau_L P_K Q_K] + \alpha_L \frac{P_L}{\tau_L}Q_L + \alpha_K \tau_L P_K Q_K}{(\frac{P_L}{\tau_L})^{\mu_L} (\tau_L P_K)^{\mu_K}} \quad (18)$$

Case C: Diversified before and specialised after PTA

In our final case, we assume that the SOE was diversified before the PTA and specialises in L-goods after the PTA. That is, $\tau_H > \tau^* > \tau_L$.

In this scenario, the SOE ceases domestic production of K-goods and becomes an importer, as the cost of purchasing K-goods from the large economy (including transport costs, τ_L) is lower than producing them domestically. This shift has uneven effects across agents: workers continue to benefit from exporting L-goods, while capitalists, though also gaining from L-good exports, experience a reduction in income due to a loss of R in their income equation⁶. Since K-goods (capital-intensive), are no longer produced domestically, the return to capital declines as the economy fully specialises in L-goods.

Similarly to Case B, the welfare (utility) of workers increases from this higher income. Capitalists experience

⁶Capitalists lose out on R in their income equation as they are no longer producing K-goods, but still maintain some R because we assume there is some capital involved in producing L-goods.

a (much larger) fall in welfare from their reduction in income, which equations (19), (20), (21) and (22) summarise.

Utility of workers and capitalists **before** PTA:

$$U_W = \frac{(1-x)[(1-\alpha_L)\frac{P_L}{\tau_H}Q_L + (1-\alpha_K)\tau_H P_K Q_K]}{(\frac{P_L}{\tau_H})^{\mu_L}(\tau_H P_K)^{\mu_K}} \quad (19)$$

$$U_C = \frac{x[(1-\alpha_L)\frac{P_L}{\tau_L}Q_L + (1-\alpha_K)\tau_L P_K Q_K] + \alpha_L \frac{P_L}{\tau_L}Q_L + \alpha_K \tau_L P_K Q_K}{(\frac{P_L}{\tau_L})^{\mu_L}(\tau_L P_K)^{\mu_K}} \quad (20)$$

Utility of workers and capitalists **after** PTA:

$$U_W = \frac{(1-x)(1-\alpha_L)\frac{P_L}{\tau_L}Q_L}{(\frac{P_L}{\tau_L})^{\mu_L}(\tau_L P_K)^{\mu_K}} \quad (21)$$

$$U_C = \frac{x(1-\alpha_L)\frac{P_L}{\tau_L}Q_L + \alpha_L \frac{P_L}{\tau_L}Q_L}{(\frac{P_L}{\tau_L})^{\mu_L}(\tau_L P_K)^{\mu_K}} \quad (22)$$

Therefore, as in Case B, the PTA will only be approved when $(1-x) > x$ i.e. the share of workers is higher than the share of capitalists as more than half of the population experience an increase in income and welfare. The inverse occurs if $(1-x) \leq x$.

2. What is the optimal redistribution from the point of view of the government?

To find an optimal redistribution from the point of view of the government, we consider transfers that will satisfy the following conditions:

1. The PTA is approved.
2. The welfare of workers is maximised.
3. The capitalists cannot be poorer than the workers.

We evaluate each of our cases to find the optimal redistribution.

The total welfare of workers and capitalists is shown by the following general utility functions:

$$U_W = \frac{(1-x)WL}{(\frac{P_L}{\tau_L})^{\mu_L}(\tau_L P_K)^{\mu_K}} \quad U_C = \frac{WxL + RK}{(\frac{P_L}{\tau_L})^{\mu_L}(\tau_L P_K)^{\mu_K}} \quad (23)$$

The welfare of an individual worker and capitalist is shown by the following general utility functions:

$$U_{W_{individual}} = \frac{W}{(\frac{P_L}{\tau_L})^{\mu_L}(\tau_L P_K)^{\mu_K}} \quad U_{C_{individual}} = \frac{W + \frac{RK}{x}}{(\frac{P_L}{\tau_L})^{\mu_L}(\tau_L P_K)^{\mu_K}} \quad (24)$$

Case A: Specialised before and after the PTA

In this case, the PTA will be approved regardless of the population distribution between capitalists (x) and workers $(1-x)$, since, as demonstrated in Question 1, welfare increases for both groups. However, to align with the government's objective of maximising workers' welfare, a redistributive policy is necessary. We show that the referendum passes when i) the majority of the population are workers and ii) the majority of

the population are capitalists.

i) When workers are the majority of the population ($x < 0.5$):

Without redistribution, capitalists' incomes—and therefore their welfare—would grow disproportionately relative to workers. The government should distribute RK among all citizens because concentrating the return on capital solely within the capitalist income equation leaves unrealized potential gains in workers' welfare. By reallocating the return on capital across all workers (L), overall worker welfare is maximised, while still ensuring that capitalists remain no worse off than workers.

This still ensures the PTA is passed because welfare increases for both workers and capitalists. After redistribution workers' and capitalists' incomes are the same at $W + \frac{RK}{L}$.

$$U_{W_{individual}} = \frac{W + \frac{RK}{L}}{(\frac{P_L}{\tau_L})^{\mu_L} (\tau_L P_K)^{\mu_K}} \quad U_{C_{individual}} = \frac{W + \frac{RK}{L}}{(\frac{P_L}{\tau_L})^{\mu_L} (\tau_L P_K)^{\mu_K}} \quad (25)$$

ii) When capitalists are the majority of the population ($x \geq 0.5$):

For a majority to vote in favour of the PTA, the welfare of the capitalists must increase. Given that the capitalists still see an increase in income and welfare from their increase in W , we are still able to redistribute the return on capital across all workers to maximise worker welfare while ensuring that the capitalists are not poorer than the workers and they vote for the referendum. We assume that the loss in return to capital is offset by the additional gain in labour wage.

Case B: Diversified before and after the PTA

Whether a redistribution occurs depends again on the proportion of the population that are made up by capitalists.

i) When workers are the majority of the population ($x < 0.5$):

As discussed in Part 1, if workers outnumber capitalists, i.e. $(1 - x) > x$, the welfare of strictly only workers would need to rise for the PTA to pass, which is the case with increased trade/lower transport costs. The government's optimal redistribution maximises workers' welfare while ensuring capitalists are not poorer than workers. It redistributes the post-PTA return to capital equally across the population, giving each individual an income of $WL + \frac{RK}{L}$. This satisfies the constraint and secures PTA approval, even if capitalists may experience a fall in income or utility.

ii) When capitalists are the majority of the population ($x > 0.5$):

In Part 1, we found that, without redistribution, capitalists oppose the PTA since their welfare falls as domestic K-production declines. However, the PTA can still pass if the government transfers part of workers' gains to capitalists.

Let workers' gain be $\Delta((1 - x)WL)$ and capitalists' loss be $\Delta(xWL + RK)$. A transfer T satisfying $|\Delta((1 - x)WL)| > T > |\Delta(xWL + RK)|$ compensates capitalists while leaving workers better off. After redistribution, capitalists earn $xWL + RK + T$ and workers $(1 - x)WL - T$. Both groups gain relative to

their pre-PTA state, and the PTA passes. This policy maximises workers' welfare under a capitalist majority by enabling trade that raises total income.

Case C: Diversified before and specialised after the PTA

To recap, in this scenario, the SOE cease production of K-goods and so capitalists lose a large portion of their capital income. As before the optimal redistribution will depend on the share of agents in the economy.

i) When workers are the majority of the population ($x < 0.5$):

In this case, the PTA is passed as workers, the majority of the population, vote for the alternative that offers them the highest welfare. Again, despite workers already being better off, it is still the government's goal to maximise the welfare of workers, and so an additional redistribution can be found to satisfy this target. Similarly to cases A and B, this can be achieved by redistributing the capital income of capitalists across the total population, while still ensuring its binding restriction is met. The proportion of the redistribution, RK , will be the same. In absolute terms, however, the redistribution here will be less than in case B as capitalists have a lower capital income. Overall capitalists are not poorer than workers.

ii) When capitalists are the majority of the population ($x > 0.5$):

A redistribution can be designed so that capitalists experience greater welfare, thereby securing majority support for the PTA's approval. Similar to case B, we optimally redistribute a certain amount, T , that lies between the income gain of workers and income loss of capitalists:

$$|(\Delta((1-x)WL))| > T > |\Delta(xWL + RK)| \quad (26)$$

Again, under these conditions, this redistribution ensures that workers and capitalists have higher income and welfare than before the PTA, meaning that capitalists vote for the PTA and the government's goal of maximising workers' welfare is sustained.

3. Conclusion

The analysis shows that signing the PTA increases overall welfare in the Small Open Economy, but the distribution of gains varies depending on the factor ownership structure. When workers are the majority, the PTA passes without opposition and redistributing capital income across the population maximises workers' welfare without making capitalists worse off. If capitalists are the majority, a carefully designed redistribution from workers to capitalists is needed to secure approval, ensuring both groups gain compared to the pre-PTA state.

Additional Notes

- **Assumptions and their limitations:** This analysis is conducted within a partial equilibrium framework, modeling a small open economy (SOE) and assessing the potential effects of a PTA while taking world prices as given. The analysis omits general equilibrium effects, cross-country capital movements, and potential feedback from changing world prices. The results are constrained by simplifying assumptions. Be aware that relaxing these may alter outcomes for both L- and K-industries and welfare for their respective factor owners.
- **Specialised Case Insight:** In a fully specialised economy, the PTA promotes more trade and specialisation. However, capitalists may gain more by demanding production of K-goods rather than relying solely on L-goods specialisation, as they would earn both wage income and returns on capital in the K-sector. This creates a potential incentive for capitalists to shift toward diversification.
- **Redistribution Objective:** The problem set assumes that the government's goal is to maximise workers' welfare rather than the welfare of capitalists or both sets of agents. As such, the redistribution policy is not designed to achieve a Pareto improvement, but to enhance worker welfare specifically.